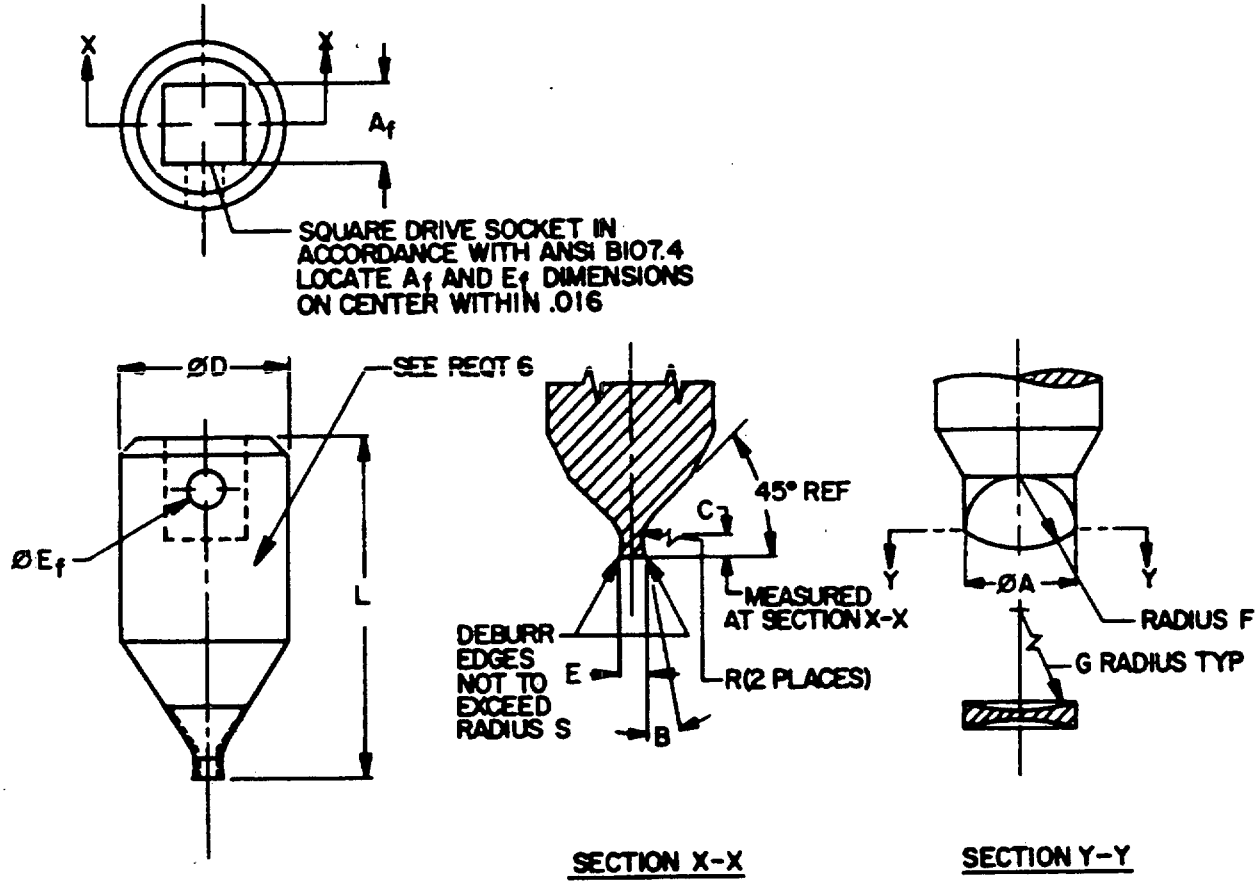
MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, HI-TORQUE, SQUARE, INTERNAL DRIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the bits described herein shall consist of this specification and the latest issue of MIL-B-9946.



NOTES:
1. All dimensions are in inches
2. Bit, socket and tip point shall be in accordance with TABLE I.

FIGURE 1. BIT, SCREWDRIVER, HI-TORQUE, SQUARE, INTERNAL DRIVE

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	DRIVER SIZE	RECESS SIZE	SCREW SIZE NOM (REF)	ØA MAX	ØA MIN	B ±1°	C +.010 -.000	E MAX	E MIN	F RAD +.000 -.010	G RAD MAX	G RAD MIN	R RAD ±.010	S RAD MAX	A_f NOM	ØD	$ØE_f$ MIN	LENGTH ±.016	ULTIMATE TORQUE MIN LBF - IN	FATIGUE TORQUE LBF - IN
0	0	0	#0, #1, #2	.172	.162	4°	.030	.0300	.0280	.115	1.500	1.375	.095	.003	.250	.500	.108	1.000	—	—
1	1	1	#3, #4, #5, #6	.279	.269	4°	.040	.0500	.0480	.165	1.500	1.375	.095	.003	.250	.500	.108	1.000	35	14
2	2	2	#8	.300	.290	8°	.053	.0600	.0575	.187	1.500	1.375	.095	.005	.250	.500	.108	1.000	65	26
3	3	3	#10	.382	.372	9°	.045	.0600	.0575	.281	2.000	1.875	.095	.005	.250	.500	.108	1.000	100	40
4	4	4	1/4	.502	.492	9°	.059	.0740	.0710	.437	3.000	2.750	.095	.005	.250	.500	.108	1.000	250	100
4A	4	4	1/4	.502	.492	9°	.059	.0740	.0710	.437	3.000	2.750	.095	.005	.375	.750	.204	1.250	250	100
5	5	5	5/16	.634	.624	9°	.069	.0800	.0770	.500	4.000	3.750	.095	.007	.375	.750	.204	1.250	500	200
6	6	6	3/8	.752	.742	9°	.093	.0960	.0930	.625	4.000	3.750	.095	.007	.375	.750	.204	1.250	675	270
7	7	7	7/16	.889	.879	9°	.106	.1130	.1095	.687	5.000	4.750	.182	.007	.500	1.000	.218	1.625	1,375	550
8	8	8	1/2	1.002	.992	9°	.135	.1430	.1395	.750	5.000	4.750	.182	.007	.500	1.000	.218	1.625	2,125	850
9	9	9	9/16	1.151	1.131	9°	.143	.1630	.1595	.875	6.000	5.750	.182	.010	.500	1.250	.218	1.625	3,000	1,200
10	10	10	5/8	1.260	1.240	9°	.150	.1830	.1795	1.000	6.000	5.750	.182	.010	.500	1.250	.218	1.625	4,050	1,620

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REQUIREMENTS:

1. Material: Alloy Tool Steel in accordance with American Society for Testing Material (ASTM) A681, Shock-Resisting Steel, Type S-2, (UNS T41902).

2. Hardness: Macrohardness, (Rockwell C Scale) 56-62HRC. Three macrohardness readings shall be taken at the location shown in FIGURE 2. Each hardness value must be within the range specified.

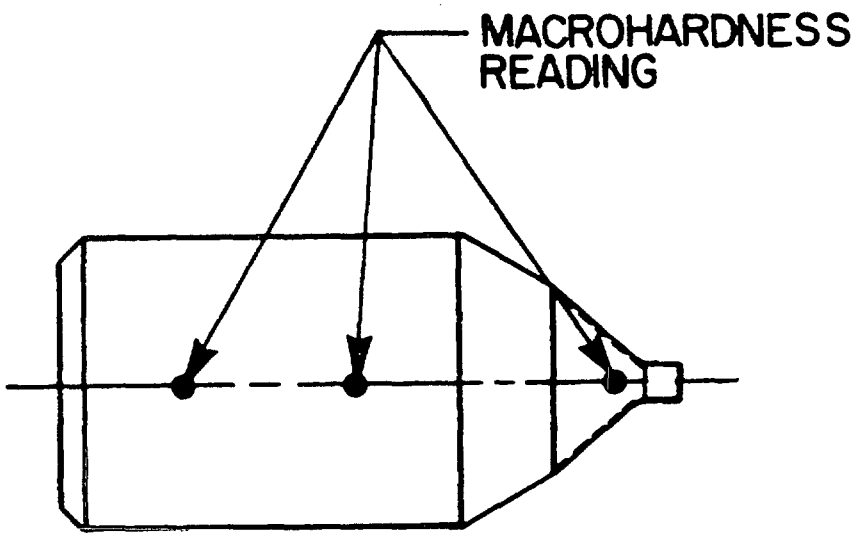


FIGURE 2. MACROHARDNESS READING LOCATIONS

Superficial hardness, (Rockwell HR15N scale) 87-90. Superficial hardness readings shall be taken on bid samples at two locations on sectionalized specimens along the blade.

3. Tip point: Tip point shall be in accordance with FIGURE 1 and TABLE I.

4. Driver bit gage: Driver bit gage shall be in accordance with MS33750.

5. Test block: Test block shall be in accordance with FIGURE 3 and TABLE II and is used to conduct torque tests as described herein and in MIL-B-9946.

6. Identification marking: Driver bits shall be marked by etching, stamping or engraving onto each bit the following minimum information:

   a. Military specification part number.

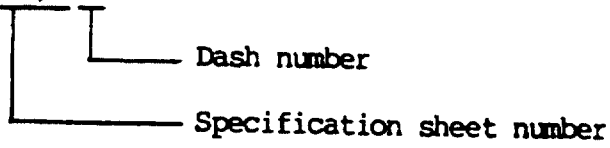
   b. Manufacturer's symbol or code.

   c. Country of manufacture.

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7. Part number: The part number consists of the letter "M", the basic number of this specification sheet, followed by a dash number taken from TABLE I.

Example: M9946/1-5



M9946/1-5 indicates - Bit, Screwdriver, Hi-Torque, Square, Internal Drive; Driver Size 5; Recess Size 5; Screw Size Nom 3/16; Square Size Nom .375; Length 1.250.

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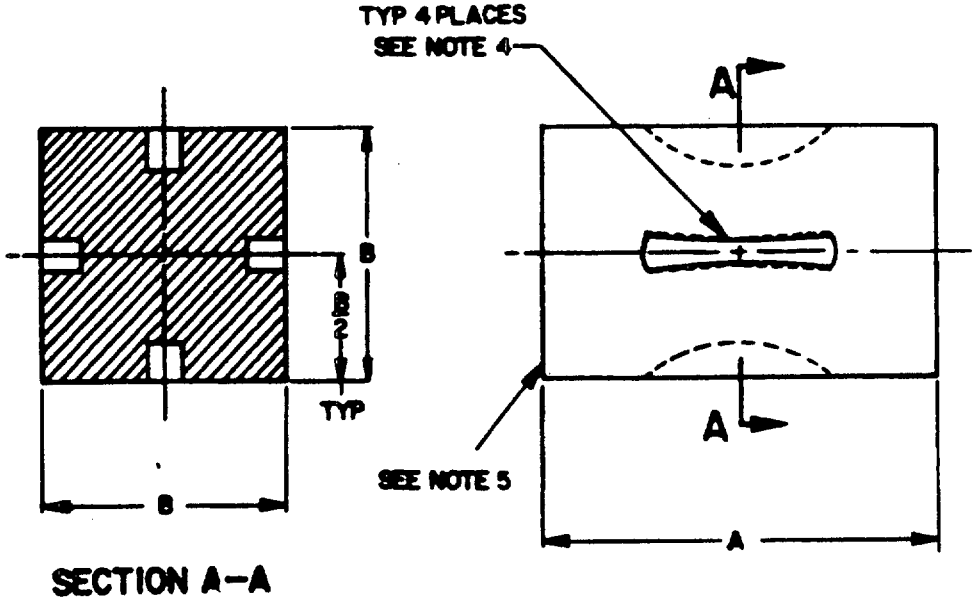


FIGURE 3. TEST BLOCK

TABLE II. TEST BLOCK DIMENSIONS FOR HI-TORQUE BITS

ADAPTER TEST BLOCK PART NUMBER	RECESS SIZE	A ±.01	B ±.03
2	2	1.00	2.50
3	3	1.00	2.50
4	4	1.00	2.50
5	5	1.00	2.50
6	6	1.00	2.50
7	7	1.00	2.50
8	8	1.00	2.50
9	9	2.00	3.50
10	10	2.00	3.50

NOTES:
1. All dimensions are in inches.
2. Material: Alloy Tool Steel in accordance with ASTM A681, Type S-2 (UNS T41902).
3. Hardness: 62-64HRC
4. Hi-Torque recesses in accordance with MS33750, 100° Flush Head Recess.
5. Mark part number on block end (method of marking optional).

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8. Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - AS
Air Force - 99

Review Activities:
Army - AV
Air Force - 80
DLA - IS

User Activities:

Civil Agencies Coordinating Activities:
GSA - FSS

Preparing Activity:
Air Force - 84

Agent:
DLA - IS

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